

JOSÉ FRANCISCO URDANETA

Technology Director · AI Architect · Full-Stack Engineer

Agentic AI · RAG · Digital Lending · Google Cloud · Engineering Leadership

Barranquilla, Colombia · Available remote across LatAm and US Eastern Time

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PROFESSIONAL SUMMARY

Software engineer with 20+ years building and leading mission-critical platforms across banking, fintech and digital lending in Latin America. I have been writing code since 2006: I started in .NET and Web Services against SQL Server, shipped native mobile banking to the Play Store and App Store, and today I build AI systems in Python and React on Google Cloud.

I direct PulseAI, 4Told Fintech's agentic AI platform for regulated finance and insurance: multi-agent architecture with business-rule reasoning, RAG and small language models, in a domain where a plausible answer is not enough because it has to be verifiable and traceable to the regulator.

I combine technical direction with hands-on execution: I design the architecture, write the code, build the team and take it to production. My work is measured by two numbers that usually move in opposite directions: I scaled credit operations by 82% while cutting support tickets by 83% in the same cycle.

CORE COMPETENCIES

Agentic AI · Multi-agent orchestration · RAG · LLMs (OpenAI, Gemini, DeepSeek) · SLM · Prompt engineering

Software architecture · Microservices · Cloud (Google Cloud, AWS) · Vertex AI · Python · FastAPI · React · .NET · Java

Digital lending · Loan origination · Risk engines · Credit bureaus · Digital onboarding · Biometrics · OCR · E-signature

Technology operations leadership · Cross-functional team management · Agile methodologies · SLA governance · Production delivery · Enterprise negotiation

TECHNICAL SKILLS

AI and intelligent systems: Agentic AI, multi-agent orchestration, multi-engine LLM integration (OpenAI, Gemini 2.5 Flash, DeepSeek), SLM strategies, prompt engineering, fine-tuning, AI cost estimation and optimization, Drools business rules engine.

RAG and knowledge engineering: ChromaDB, vector search with HNSW indexing and cosine similarity, text-embedding-004 embeddings, structural chunking with metadata propagation, LangChain text splitters, domain ontologies and taxonomies, retrieval evaluation (Precision@k, Recall@k, MRR), grounding and citation validation.

Cloud and infrastructure: Google Cloud Platform, Vertex AI, Cloud Run, Firebase, Firestore, Firebase Auth, AWS, Supabase, Railway, Docker, GitHub Actions, distributed systems, resource and cost governance.

Backend: Python, FastAPI, Uvicorn, Pydantic, Java, .NET / C#, Node.js, microservices architecture, REST API design and integration, Web Services, BPM.

Frontend and mobile: React, Vite, Angular, JavaScript, Tailwind CSS, HTML5, CSS3, PWA, native Android (Android Studio), native iOS (Xcode), Play Store and App Store releases.

Databases and analytics: SQL Server, PostgreSQL, Firestore, ChromaDB, data modelling, data lakes, ETL, Power BI, Pandas, descriptive analytics, operational KPIs.

Fintech domain: Digital lending, loan origination, automated risk engines, credit bureau integration, digital onboarding, biometric validation, OCR, documentary verification, electronic signature, core banking, mobile banking.

Leadership and process: Technology operations direction, leadership of distributed cross-functional teams, Scrum and Kanban, stakeholder management, architectural decision negotiation, production reliability and SLA governance.

PROFESSIONAL EXPERIENCE

4TOLD FINTECH | Colombia (remote) | April 2017 - Present

Director of Operations (2020 - Present)

Director of Digital Platforms (2018 - 2020)

Mobile Technical Lead (2017 - 2018)

B2B digital lending fintech: origination, onboarding and risk-decisioning platforms for retail and regulated financial institutions in Colombia and Peru.

- Direct PulseAI, the company's agentic AI platform: multi-agent architecture (receptionist agent, intelligent orchestrator and domain-specialist agents) with an inferential business-rules layer in Drools governing every decision for regulatory traceability.
- Governed the multi-engine LLM architecture integrating OpenAI, Gemini and DeepSeek behind a single orchestration layer, defining routing strategy, contract abstraction and provider independence at architectural level.
- Designed RAG and SLM strategies to migrate workloads to faster, cheaper models; estimated AI spend and governed compute provisioning, owning the cost and latency trade-off across the stack.
- Defined the governance controls separating generative conversation from business decisioning, enabling compliant inbound and outbound operation under consent and contact rules, a hard requirement in regulated markets.
- Integrated the agents omnichannel across CRM, core banking, mobile banking, web and WhatsApp, with a 12-week agile roadmap for the agentic pilots.
- Scaled digital credit operations by 82% (453 to 825 operations per month) while cutting support tickets by 83% (208 to 36 per month) in the same cycle, then sustained growth to 889-964 operations per month with tickets at record lows.
- Implemented and took to production 10 clients across Colombia and Peru, including national retail, agricultural development banking, municipal savings banks, banks and credit unions.
- Cut client portfolio delinquency by 5% with automated risk engines wired to credit bureaus, lifted their loan placement by 20% and reduced their costs by 20% through digital transformation of approval flows.
- Reduced operating costs by 30%, support response times by 40% and man-hours by 20%; delivered 25% more projects per year on the same budget.
- Automated digital onboarding end to end: biometric validation, OCR, documentary verification and electronic signature, orchestrated across third-party services under regulatory constraints.
- Led a 20-person cross-functional team (backend, web, mobile, rules engine, client-facing implementation, PM and technical support) under Scrum and Kanban across multiple countries and client tiers.

VEDANA SOLUTIONS | LatAm (remote) | December 2022 - December 2025

Technical Director

Product, data and operations consulting for companies across Latin America.

- Designed and implemented the Orienta and Loanware data-lake analytics products, architecting ingestion, retrieval and AI-driven analysis to turn enterprise operational data into decision-making capability.
- Owned the full product cycle: concept, architecture, development kickoff, UAT, market assessment, risk analysis, positioning, trademark registration and sales, negotiating technical scope directly with stakeholders.
- Diagnosed processes, identified bottlenecks and designed solutions on constrained budgets, building custom software as the supporting tool.

B.O.D. BANCO UNIVERSAL | Venezuela | May 2010 - December 2016

Mobile Development Manager (2014 - 2016)

BPM Development Coordinator (2010 - 2014)

Universal bank with high-volume transactional operations in a regulated environment.

- Launched the transactional digital banking platform, web and native mobile, into production: 5,000 daily active users, USD 2 million transacted per day and 20% more operations than the legacy system.
- Led a 15-person cross-functional team and shipped the applications to the Play Store and App Store against enforced quality standards.
- Coordinated BPM development: feasibility analysis, process discovery, estimation and resource allocation, and rollout of new technologies.

PYRAMID CONSULTING | Madrid, Spain | 2014 - 2015

Project Manager / Team Leader

- Led native mobile and responsive web development projects for clients in Spain.

NEWTECH SISTEMAS | Venezuela | 2006 - 2010

Development Consultant

- Built .NET applications and Web Services against SQL Server for corporate clients.

AI AND PRODUCT PORTFOLIO

PulseAI (4Told Fintech). Production agentic AI platform for regulated finance and insurance. Multi-agent architecture, multiple LLM engines behind one orchestration layer, inferential reasoning with Drools, RAG and SLM. Role: product and architecture direction.

Zona Barrica (client). Cloud-native whisky collection and tasting platform with an AI Sommelier. The engine builds a palate DNA from a 14-attribute sensory profile and generates personalised recommendations with Gemini 2.5 Flash on Vertex AI. Four FastAPI microservices on Cloud Run, React frontend, Firestore, with a configurable rate limit on AI generation to bound per-user cost. Role: architecture, full-stack, AI and infrastructure. <https://zonabarrica-1541.web.app/>

RunCoach AI (applied research, MSc in ICT). Redesign from a simple agent into a knowledge-based system: domain ontology and taxonomy, structural chunking with metadata propagation, 768-dimension text-embedding-004 embeddings and a ChromaDB vector store with HNSW indexing and cosine similarity. The agent synthesises the query, retrieves top-k with metadata filtering and validates citations before answering. Precision@3 of 0.815, Recall and MRR of 0.944, citation precision of 1.000 over an 18-case golden set. <https://runcoach-ai-496921.web.app/>

Digital Onboarding (4Told Fintech). Digital identity and anti-impersonation enrolment for credit origination: biometric validation, OCR, documentary verification and electronic signature, wired into risk engines and credit bureaus. In production for retail and financial institutions in Colombia and Peru.

Los Líderes Encomiendas (client). Cross-border Colombia-Venezuela logistics operation digitised end to end: a multi-role PWA for operations (mailboxes, tracking, roles and reporting) and a commercial landing optimised for conversion and SEO, deployed separately. React 19, Supabase, Tailwind CSS, Railway. <https://www.loslideresencomiendas.com/>

Orienta and Loanware (Vedana Solutions). AI analytics over a data lake for enterprise decision-making, covering the full product cycle from concept to trademark registration.

jutechnologies.com (personal project). Bilingual portfolio built in React and Vite with no UI framework, a custom design system and animated SVG visuals. Deployed on Railway. Source at github.com/jfurdaneta/jutechnologies. <https://www.jutechnologies.com>

EDUCATION

M.Sc. in Information and Communication Technologies, Universidad del Zulia. In progress, expected 2027.

B.Sc. in Computer Science, Universidad del Zulia. 2007.

Current research: intelligent agents and RAG systems applied to training prescription, grounding a production system in peer-reviewed literature.

LANGUAGES

Spanish (native) · English (B2, professional working proficiency)